

Discrete Mathematics Problem Sheet 2

Feb 25, 2026

1. Prove bijectively that

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2.$$

2. Let a_n denote the number of ways to write n as a sum of distinct positive integers, and b_n be the number of ways to write n as a sum of positive odd integers. Show bijectively that $a_n = b_n$ for all n . Can you show it via generating functions as well?
3. Let s_n denote the number of subsets of $[n]$ with no two consecutive numbers in it. How is s_n related to Fibonacci numbers?
4. There are n parking spots marked 1 to n and the sequence $(a_1, a_2, \dots, a_n)$ are the preferred slots for n cars (numbered 1 to n), where a_i is the preferred spot for car i . Cars arrive in order, and each car tries to park in its preferred spot. If it is occupied it parks in the next free spot. If there is no free spot ahead for a car, that is a *failure*. The sequence is a valid *parking function* if there is no failure. Show that the number of valid parking functions is $(n+1)^{n-1}$ by giving a bijection to labeled trees on $n+1$ vertices.
5. Let w be any binary string of length k . For $n \geq k$, count the number of binary strings of length n that do not contain w as a *subsequence*.
6. Show that the n^{th} Catalan number $\frac{1}{n+1} \binom{2n}{n}$ counts the binary strings of length $2n$ that do not contain any of the following strings as *subsequences*: $1^{n+1}, 1^n 0, 1^{n-1} 0^2, \dots, 1^i 0^{n+1-i}, \dots, 10^n, 0^{n+1}$.
7. Solve the following recurrence relation by relating it to a problem solved in class (or otherwise):

$$a_0 = 1$$

$$a_n = n a_{n-1} + (-1)^n \text{ for } n \geq 1.$$

8. Give a combinatorial proof for the principle of inclusion exclusion

$$|\cap_{i=1}^n \overline{A_i}| = \sum_I (-1)^{|I|} |A_I|$$

by first moving all the negative terms in the summation to the LHS so that the equation assumes the form $A = B$, where both A and B are now sums of positive terms. Then define suitable sets whose sizes are these positive terms and give a bijective correspondence that will prove the equation.

9. Solve the recurrence relation $a_r + 3a_{r-1} + 2a_{r-2} = f(r)$ where $f(r) = 1$ for $r = 2$ and $f(r) = 0$ otherwise. Assume the boundary conditions $a_0 = a_1 = 0$.
10. Gossip is spread among r people via telephone. In a phone call between two people A and B , they exchange all the gossip they have heard so far. Let a_r denote the minimum number of phone calls so that all the gossip will be known to everyone. Show that $a_2 = 1$, $a_3 = 3$ and $a_4 = 4$. Then show that $a_r \leq a_{r-1} + 2$. Finally, show that for $r \geq 4$, $a_r \leq 2r - 4$.
11. Let $G(x)$ be the exponential generating function (egf) for the number of labeled n vertex graphs and $C(x)$ be the egf for the number of connected labeled n vertex graphs. Show that $G(x) = \exp(C(x))$.
12. Let a_n denote the number of permutations on $[n]$ with only 1-cycles and 2-cycles. Derive its egf and find a closed form for it. Does a_n have a different combinatorial meaning?
13. Derive the generating function for the number of length- n words over a k -letter alphabet such that no letter occurs consecutively in the word.